Energy Data Analysis & Dashboard Project Documentation

1. Project Overview

This project focuses on analyzing energy consumption and production data from 3 main sources:

- **ESB (Electricity Supply Board)** - Imported consumption data.

- **SolisCloud** - Solar inverter and PV system data.

- **Meteostat API** - weather data.

The goal of the project was to build a complete end-to-end data pipeline: from data ingestion, cleaning, feature engineering, exploratory analysis, calculation of key performance indicators (KPIs), generating visualizations, and publishing the results to a dashboard in **Google Looker Studio**.

This documentation serves as a portfolio piece that can be presented during job interviews to demonstrate skills in Python, pandas, data transformation, analytics, and dashboard creation.

2. Data Sources

2.1 ESB Data

- Source: ESB customer portal downloads - 30 min readings in calculated kWh
- Format: CSV files
- Contains: Daily consumption, timestamps, meter readings, import/export data

2.2 SolisCloud Data

- Source: SolisCloud customer portal download (monthly data)
- Format: XLS
- Contains: PV production, inverter metrics, battery charge/discharge, grid import/export, load consumption

2.3 Meteostat Data

- Source: API
- Contains: date, min temp, max temp, the daily sunshine total minutes and the daily precipitation

3. Tools & Technologies

- **Python** (pandas, matplotlib, rest api)
- **Jupyter Notebook**
- **Google Looker Studio (formerly Data Studio)**
- **Excel (via pandas ExcelWriter)**
- **XLS/ CSV ingestion**

4. Data Pipeline

Step 1: Data Ingestion

- Imported multiple CSV/XLS files for both ESB and SolisCloud.
- Used pandas to:

    - Load large datasets efficiently
    - Standardize column names
    - Convert timestamps into proper datetime formats

Step 2: Data Cleaning & Preparation

Performed data quality checks: - Checked and handled null values - Identified and removed duplicates - Fixed inconsistent date formats - Converted numeric fields into proper types (float, int)

Step 3: Feature Engineering & KPI Calculation

Generated several key indicators, including: - **Daily PV production (kWh)** - **Load consumption** - **Energy imported from grid** - **Energy exported to grid** - **Energy used directly from solar (self-consumption)** - **Battery charge/discharge flows** - **Solar vs Grid % breakdown**

Used custom formulas to correctly calculate PV → Load direct usage without double-counting battery flows.

Step 4: Data Transformation for Visualization

- Created new data points in order to differentiate between night and day tariffs based on timestamps.
- Reshaped data using pivot_table and melt
- Generated summary tables grouped by:
    - Day
    - Month
- Prepared datasets suitable for Looker Studio ingestion
- Ensured all KPIs were pre-calculated for dashboard performance

Step 5: Exporting Results

- Exported processed tables to structured Excel files using pandas ExcelWriter
- Ensured timestamps were properly formatted
- Created multiple sheets: raw data, cleaned data, KPIs, visual data, summaries

Step 6: Dashboard Integration

Published the processed data to **Google Looker Studio**, where I designed a complete interactive dashboard:

- Time-series charts for consumption, PV production, grid import/export

- Breakdown charts (day vs night tariff)

- Daily/monthly summaries

- Monthly trends comparison with previous years

- Custom solar vs consumption Sankey-style visuals where possible

5. Exploratory Data Analysis (EDA)

Performed EDA to understand patterns:

- PV production patterns by day/month

- Load consumption distribution

- Grid dependency trends

- Battery behavior (charge/discharge cycles)

- Weather correlation

6. Key Insights & Findings

Examples of insights derived:

- Peak daily PV production times

- Seasonal variability of solar generation

- Self-consumption rate (solar used directly vs exported)

- Times of day with high grid dependency

- Battery efficiency patterns

7. Challenges & Solutions

Time format inconsistencies (YYYY-00-DD issue)

Solved through correct `to_datetime` parsing and setting proper day/month formats.

When exporting the data into excel, there is an issue when excel exports datatime so using python I formatted the dates for each table automatically.

Day vs Night Tariffs

The csv/xls files from the ESB platform does not provide a breakdown of the smart tariff since this is provided by Energia. I had to create new datapoints to capture this information and transform the data using melt method in order to create specific charts like donut to compare day vs night %

Create Derived Extra Derived Metrics

ESB platform is not the main provider so I had to adjust the cost and export KPIs based on the tariff mentioned above to have the most accurate information.

API certificate validation issues

Handled via proper SSL settings or switching to alternative API methods.

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8. Future Improvements

- Add forecasting models (ARIMA, Prophet)
- Build a full ETL pipeline using Airflow or Prefect

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9. Skills Demonstrated

- Data Cleaning & Engineering
- Exploratory Data Analysis
- Python (pandas) proficiency
- Data visualization design
- Dashboard building
- Problem-solving of real-world messy data

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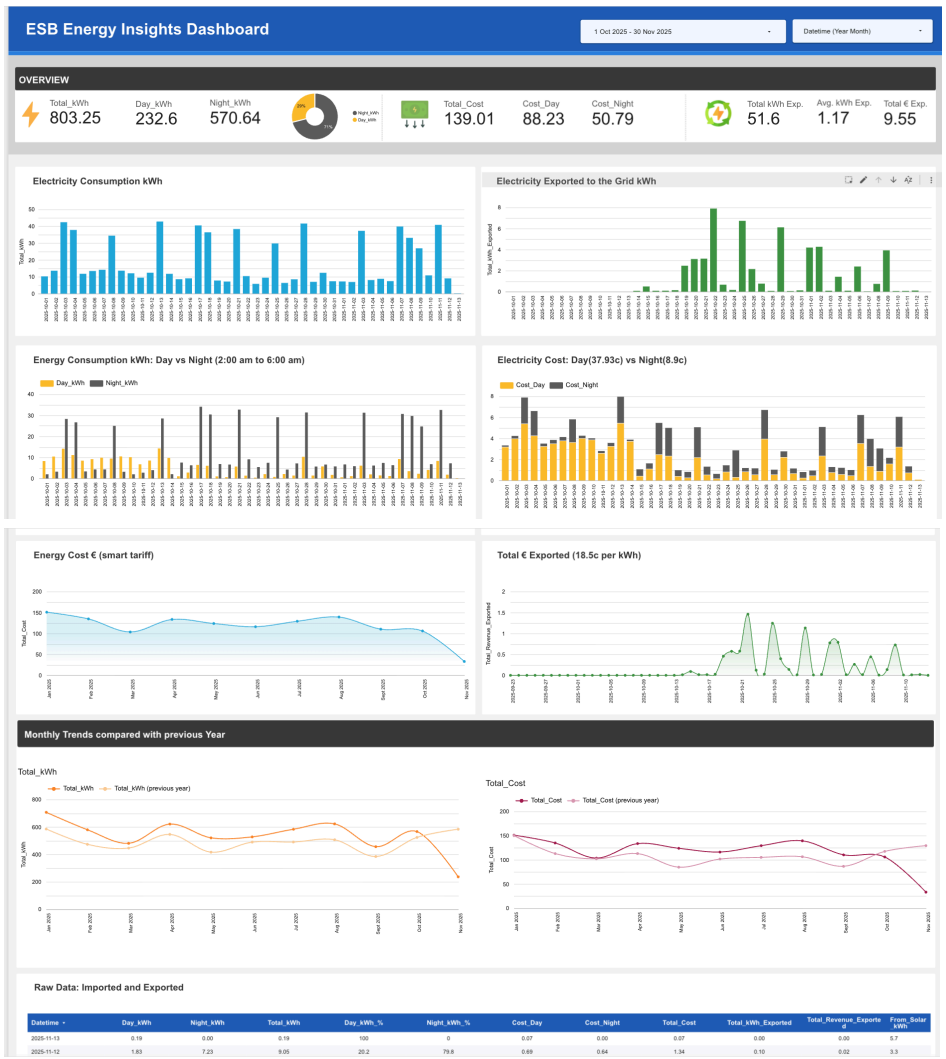
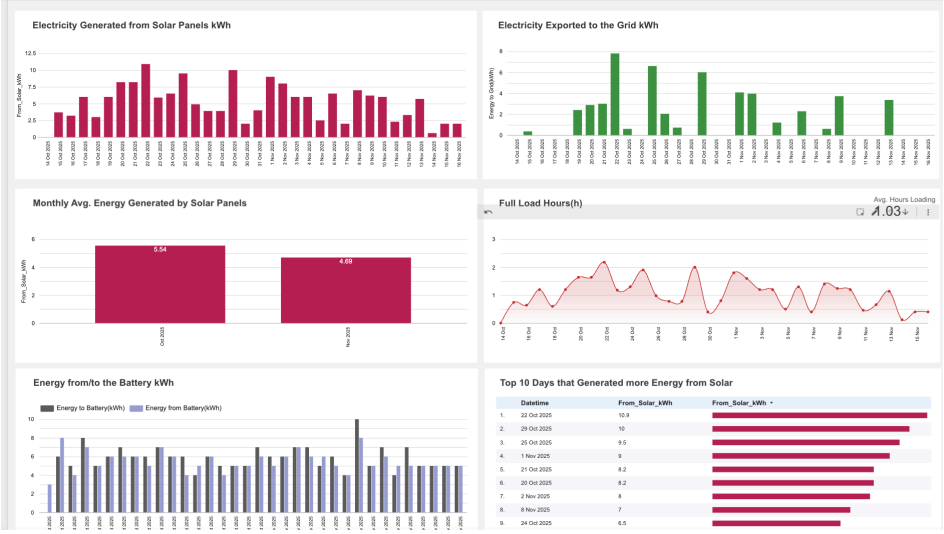
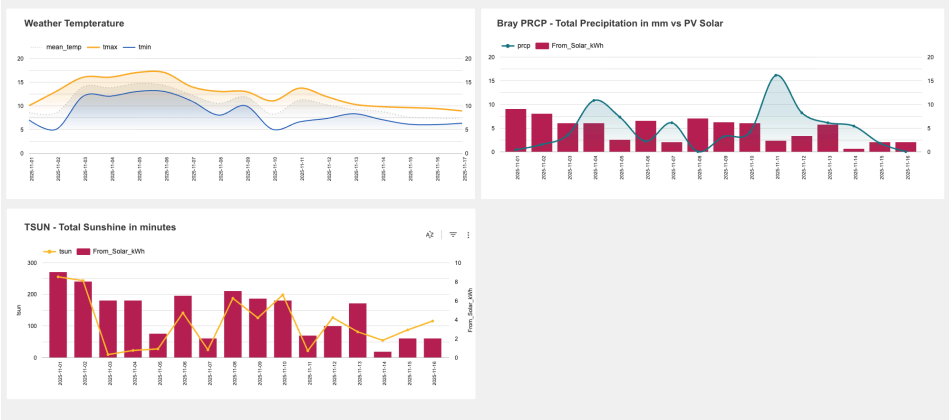
10. Appendix

- Example rawdata (link)
- Example code snippets (to be added)
- Sample visualizations




Electricity Generated from Solar Panels kWh
From_Solar_kWh
Electricity Exported to the Grid kWh
Energy to Grid(kWh)
Monthly Avg. Energy Generated by Solar Panels
From_Solar_kWh
6.54
4.69
Oct 2025
Nov 2025
Full Load Hours(h)
Avg. Hours Loading
4.03
Energy from/to the Battery kWh
Energy to Battery(kWh)
Energy from Battery(kWh)
Top 10 Days that Generated more Energy from Solar
Datetime
From_Solar_kWh
1. 22 Oct 2025 10.9
2. 29 Oct 2025 10
3. 25 Oct 2025 9.5
4. 1 Nov 2025 9
5. 21 Oct 2025 8.2
6. 20 Oct 2025 8.2
7. 2 Nov 2025 8
8. 8 Nov 2025 7
9. 24 Oct 2025 6.5


Weather Tempterature
mean_temp
tmax
tmin
Bray PRCP - Total Precipitation in mm vs PV Solar
prcp
From_Solar_kWh
TSUN - Total Sunshine in minutes
tsun
From_Solar_kWh